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"2G, 3G, 4G wireless networks"

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Abstract: *In this article, CDMA (Code Division Multiple Access), GSM (Global Systems for Mobile Communications), TDMA (Time Division Multiple Access), 802.11, WAP (Wireless Application Protocol) wireless technology protocol), 3G and 4G (third and fourth generation technologies), GPRS (General Packet Radio Service, data packet transmission service), Bluetooth (medium and short distance network), EDGE (Enhanced Data) Rates for GSM Evolution, improved GSM network).*

Key word: *Wi-Fi (Wireless Fidelity) - "wireless connection", WLAN (Wireless Local Area Network - Wireless Local Area Network), medium and short distance network (Bluetooth).*

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At the same time that information technologies are rapidly developing, communication and information exchange are also developing at a great speed. The role of means of communication in increasing the ease of communication is incomparable. Because the quality of communication is closely related to the means of communication. Initially, communication was carried out only through wire connectors. These wires connected users to each other through communication centers, and thus communication was established between cities and countries. Nowadays, modern products of such wires are used. An example of these is fiber optic cables. It has several advantages. But even such fiber connections cannot meet the requirements of the time. Because this wire connection has several disadvantages related to it. Examples of these include pulling these wires to some communication centers, inconveniences in the placement of wires, etc.

The history of wireless technologies of information transmission began with the first radio signal transmission at the end of the 19th century, and the appearance of amplitude modulation radio receivers in the 20s of the 20th century greatly influenced the development of these technologies. By the 1970s, the first wireless radiotelephones were created that transmitted sound through radio waves. Initially, these worked on analog networks, but in the early 80s, the GSM standard was developed, which means that the transition to digital standards has begun, which provides good spectrum distribution, the best signal quality and the best security. In the 90s of the 20th century, the processes of strengthening the state of wireless networks took place, which led to the rapid development of these technologies. Today, wireless technologies are firmly entrenched in our daily lives, while providing high speed, they provide new devices and services.

New CDMA (Code Division Multiple Access - channel code distribution technology), GSM (Global Systems for Mobile Communications - global system of mobile communication networks), TDMA (Time Division Multiple Access -

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channel distribution technology by time), 802.11, WAP (Wireless Application Protocol -wireless technology protocol), 3G and 4G (third and fourth generation technologies), GPRS (General Packet Radio Service, packet data transmission service), Bluetooth (medium and short distance network), EDGE (Enhanced Data Rates for GSM Evolution, an improved GSM network) and the diversity of similar technologies mean that a fundamental shift is beginning in this field. The development of wireless local networks (WLAN) and medium and short-range networks (Bluetooth) is very promising. Wireless local networks are widely used in airports, universities and institutes, hotels, restaurants, enterprises and organizations.

The development of wireless network standards began in 1990 with the establishment of the 802.11 committee by the worldwide IEEE (Institute of Electrical and Electronics Engineers). The worldwide spider web and the idea of working with wireless devices in this network serve as an important impetus for the development of wireless technologies. By the end of the 90s, WAP-service was provided to users. It should be noted that at the beginning this service did not arouse much interest in many people. WAP-service provided a set of news, weather, daily and other services as the main information services. Also, Bluetooth and WLAN were used very little due to the high cost of these means of communication. But the drop in prices caused an increase in demand and interest in these tools. By the middle of the first decade of the 21st century, the number of wireless Internet service users reached several tens of millions. It is in the first place with the emergence of wireless Internet.

A significant aspect of the advancement of wireless technologies is that these technologies are easily accessible to home users. With the increase in the number of home network devices, many wires connecting these devices to each other are becoming the main problem of this network. This, in turn, causes the transition to wireless technologies. Although the number of individual users of wireless

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technologies is significant, the fastest growing segment is its corporate users. Wireless data transmission is considered an important strategic tool, it ensures productivity increase in the enterprise (employees have constant and fast access to corporate information, they quickly learn about news), increases the quality of customer service (at the same time, their complaints and wishes can be received and felt at the same time), creates an advantage over competitors (increases the speed of information exchange and decision-making). In one word, we can say that wireless technologies are technologies of the future.

The problem of transferring information from one computer to another has been around since the dawn of computing. Such transfer of information makes it possible to organize the joint work of separately used computers, to solve the same problem with the help of several computers. Especially in recent versions of the Windows operating system, the availability of more advanced network tools frees you from purchasing special network software.

In addition, it will be possible to specialize each computer to perform a certain task, to use the resources of computers together, and to solve many other problems. Also, one of the main technical indicators of the network is the ability to work with a large load, that is, the speed of information exchange (in other words, with a large traffic). If the information exchange management mechanism used in the network is not effective, then computers may wait a long time for information transfer. Even if the information is transmitted at a high speed and error-free after the turn, the network user still has to wait for a certain time to use the network resources.

In order for any information transfer control mechanism to work guaranteed, the number of computers and information that can be connected to the network should be known in advance. It is natural that any mechanism will not be able to transfer information as a result of connecting more computers to the network than planned, causing an increase in the load. Finally, as a network, as in the original

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meaning of this word, it is necessary to understand such an information transmission system, which must have united several dozen local computers.

The development of wireless local networks (WLAN) and medium and short-range networks (Bluetooth) is very promising. Wireless local networks are widely used in airports, universities and institutes, hotels, restaurants, enterprises and organizations. The development of wireless network standards began in 1990 with the establishment of the 802.11 committee by the worldwide IEEE (Institute of Electrical and Electronics Engineers). The World Wide Web and the idea of working with wireless devices in this network serve as an important impetus for the development of wireless technologies. Wi-Fi technology is one of the most promising computer networks in the computer world today. Wi-Fi (Wireless Fidelity) is made up of English words and means "wireless connection". Wi-Fi technology is one of the types of sending digital data through radio channels. When this technology was created, it was first intended for corporate users, and it was predicted that it would replace the cable network. As we know, creating a computer network with a cable network requires the manual installation of several thousand cables and the installation of a special network topology. Wi-Fi is a standardized technology for wireless data exchange operating at reduced control frequencies of radio frequencies.

Usually WLAN (Wireless Local Area Network — Wireless Local Network) networks are created through the Wi-Fi network. In this network, it will be possible to see communication and information exchange through high radio waves. This system is used as an extension of the cable network or as an alternative to it, in one office, an entire building or a territory of an area. Wi-Fi technology saves you money on the expensive process of laying down thousands of cables, and the simplicity of installation saves time on complex technical installation processes, making this network superior to other networks. Due to the fact that wireless networks use radio

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frequencies, radio waves pass through walls or similar obstacles in buildings or offices in general, and nothing at all can block it (except for distance, of course!). Wireless networks are inherently more reliable than wired networks. Most WLAN networks have a range or coverage area of 160 meters, which of course depends on the nature and extent of obstacles in its path. The speed of this network can be equal to the cable network and can be several times higher than it. Of course, it depends on which standard is used.

Wi-Fi technology is one of the most promising computer networks in the computer world today. Wi-Fi (Wireless Fidelity) is made up of English words and means "wireless connection". Wi-Fi technology is one of the types of sending digital data through radio channels. When this technology was created, it was first intended for corporate users, and it was predicted that it would replace the cable network. As we know, creating a computer network with a cable network requires the manual installation of several thousand cables and the installation of a special network topology. Wi-Fi is a standardized technology for wireless data exchange operating at reduced control frequencies of radio frequencies.

Wi-Fi communication network In a WLAN network, like normal networks, the data throughput depends on its topology, loading, the distance of the loading point, and similar parameters. One of the most convenient aspects of this network is its easy installation, and the second is that there are no problems in expanding the Wi-Fi network, or in other words, it is the simplest network that is easy to expand. need To expand this WLAN network, from a practical point of view, it is enough to create new connection points. A user who buys a Wi-Fi device or router can easily assume that he has the following options:

1. A special device that works with additional devices together with multi-functional multi-service wireless communication

2. You will be able to exchange data over long distances at high speed.

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3. You don't need to do almost anything to expand the network: it is sufficient for the new user to know the network connection password to connect to the network.

4. This user will be using the latest advances in Internet technologies and telecommunications.

INTERNET is an integrated network made up of computer networks around the world, which works on the basis of a single language - template - set of rules for information exchange. The INTERNET network makes it possible to get any information regardless of time and place. The development, perfection and simplicity of the hardware and software tools have reached the level where most users can install their networks on their own. But every networker or user who wants to get acquainted may not be satisfied with the existing literature. Because most of the literature tried to cover many big issues in one book. The problem of transferring information from one computer to another has been around since the dawn of computing.

Currently, it unites all the continents of the world. Some of the computer networks in the Internet, CSNET, NSFNET, are large networks, and they themselves consist of several networks. Internet coordination is managed by the NIC (NetWork Information Center), SRI (Stanford Research Institute, often referred to as SRI-NIC) at Stanford University. ,

SMTP (Simple Mail Transport Protocol) is used for e-mail from regular mail accounts.

MCI, ICT provides mail, fax and telex services to its customers.

NSFNET is the US National Foundation's largest supercomputer-connected network of research institutions, corporations, and government agencies in the United States.

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UUNET is a non-commercial network that provides information on USENET, such as getting starter texts on UNIX.

UUCPNET (Unix to Unix Copy) is international e-mail, and data is sent through programs called UUCP.

UUCP is a protocol for transmission, a set of files for communication purposes, and a set of commands for communication programs. It is widely used for sending e-mails and participating in teleconferences.

They need to understand each other in order to communicate through computers worldwide. ITO (International Telecommunication Union) International Telecommunication Union was established in order to ensure the proportionality of computers. It consists of 3 bodies controlling telephone and data transmission systems. This body is called CCITT and develops proposals in the field of telephone, telegraph, data transmission services. International standardization ISO (Organization and Standardizations) - It unites more than 100 countries. IEEE international organization (Institute of Electrical and Electronics Engineers), besides publishing various magazines, produces many standards on electronic and computing technology.

In conclusion, it should be noted that as a result of the development of wireless communication technology, expensive wired and satellite communication systems are used less and economic savings can be achieved. By using such technologies, the user can effectively use communication exchange and other communication services in a mobile and stationary state, in any geographic environment, and gains are made in terms of time and economy. Taking this into account, we also study the undiscovered aspects of Wi-Fi, WiMax and Wi-Bro technologies, find ways to use them, optimize their parameters, and design new network topologies.

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